

JOSEPH WACHIRA MACHARIA

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Software Engineer | Computer Science Professional

PROFESSIONAL SUMMARY

Computer Science student (graduating Oct 2026) with proven software engineering experience in digital transformation projects. Built a production-ready birth and death registration system and an AI-powered educational chatbot. Proficient in Python, Java, React, Flask, and relational databases. Passionate about automating workflows and deploying scalable, AI-enhanced applications. Seeking software engineering or graduate roles where systems thinking and clean code drive operational impact.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, HTML5, CSS3

Frameworks & Libraries: React, Flask, REST APIs, Git, GitHub

Databases: MySQL, PostgreSQL

Software Engineering: OOP, UML Modeling, Agile, Software Testing, System Analysis & Design

AI & Cloud: Machine Learning Fundamentals, Intelligent Chatbots, Automation, Cloud -Based Applications

Tools: VS Code, [Draw.io](https://draw.io), Figma, MS Office

PROJECTS

Digital Birth and Death Registration System | *Final Year Project*

- Designed and developed a web-based system to digitize civil registration processes.
- Reduced manual data entry time by an estimated 40% through automated workflow scripting.
- Implemented secure authentication and role-based access control; designed relational schema supporting 10,000+ record entities with <100ms query response.
- Produced comprehensive documentation: UML diagrams, system analysis reports, and test cases.

AI Educational Assistant Chatbot | *Personal Project*

- Developing an AI-powered chatbot (Python, modern LLM APIs) that delivers contextual learning support.
- Currently handling 50+ simulated student queries per day with 85% answer accuracy.

- Designing scalable architecture to support real-time information retrieval and future AI enhancements.

Interactive Photography Portfolio Website | *Personal Project*

- Built responsive portfolio website with modern front-end practices (HTML, CSS, JavaScript).
 - Improved page load speed by 30% via lazy loading and image compression techniques.
 - Enhanced user experience through intuitive navigation and professional branding.
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PROFESSIONAL EXPERIENCE

Industrial Attachment Trainee

Office of Birth and Death Registry, Karatina, Kenya

May 2025 – August 2025

- Supported digitization and management of civil registration records.
 - Proposed and prototyped an automated data validation script, reducing manual verification time by 15 hours/week (adopted by the team).
 - Analyzed existing workflows, identified three bottlenecks, and documented processes for digitization.
 - Used SQL to identify and clean 2,000+ duplicate civil records, improving data integrity.
 - Gained hands-on experience with government information systems and digital transformation.
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ADDITIONAL EXPERIENCE

Freelance Photographer | *Self-Employed*

2021 – Present

- Manage end-to-end client projects: planning, execution, digital delivery.
 - Automated photo delivery pipeline using cloud storage and batch renaming scripts (Python).
 - Managed digital assets and metadata for 20+ clients, ensuring version control and backups.
 - Built strong client relationships through effective communication and project management.
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EDUCATION

Bachelor of Science in Computer Science

2022 -2026

Karatina University – Expected Graduation: October 2026

KCSE – Gituru Boys High School

2018 -2021

Grade: B Plain

Relevant Coursework: Software Engineering, Artificial Intelligence, Database Management Systems, Information Systems, Computer Networks, Web Development, Human-Computer Interaction

REFERENCES

Dr. Zablon Okari

Lecturer, Karatina University

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